Draft

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1 Source term analysis

1.1 Electric field source terms

Use the conventions $\delta_x^{E,2} = \delta_x^H \delta_x^E$ and $\delta_x^{H,2} = \delta_x^E \delta_x^H$. Let $S_{E,1}^n$ be the electric field source term in between n and $n+\frac{1}{2}$, and $S_{E,2}^n$ be that in between $n+\frac{1}{2}$ and $n+1$.

$$\left(1-\gamma\delta_x^{E,2}\right) E_z^{n+\frac{1}{2}} = E_z^n + C_b \delta_x^H H_y^n + S_{E,1}^n \tag{1}$$

$$H_y^{n+\frac{1}{2}} = H_y^n + D_b \delta_x^E E_z^{n+\frac{1}{2}} \tag{2}$$

$$E_z^{n+1} = E_z^{n+\frac{1}{2}} + C_b \delta_x^H H_y^{n+\frac{1}{2}} + S_{E,2}^n \tag{3}$$

$$H_y^{n+1} = H_y^{n+\frac{1}{2}} + D_b \delta_x^E E_z^{n+\frac{1}{2}} \tag{4}$$

Shifting back equations (3) and (4) one full step gives

$$E_z^n = E_z^{n-\frac{1}{2}} + C_b \delta_x^H H_y^{n-\frac{1}{2}} + S_{E,2}^{n-1} \tag{5}$$

$$H_y^n = H_y^{n-\frac{1}{2}} + D_b \delta_x^E E_z^{n-\frac{1}{2}} \tag{6}$$

To eliminate H_y from equations (1)–(3), substitute equation (2) into equation (3):

$$E_z^{n+1} = E_z^{n+\frac{1}{2}} + C_b \delta_x^H H_y^n + \gamma \delta_x^{E,2} E_z^{n+\frac{1}{2}} + S_{E,2}^n. \tag{7}$$

Equation (1) gives

$$C_b \delta_x^H H_y^n = \left(1-\gamma\delta_x^{E,2}\right) E_z^{n+\frac{1}{2}} - E_z^n - S_{E,1}^n. \tag{8}$$

Using $\gamma\delta_x^{E,2} = C_b D_b \delta_x^H \delta_x^E$, equations (7) and (8) reduce to

$$E_z^{n+\frac{1}{2}} = \frac{1}{2}\left(E_z^{n+1} + E_z^n + S_{E,1}^n - S_{E,2}^n\right). \tag{9}$$

Shifting back one full step gives

$$E_z^{n-\frac{1}{2}} = \frac{1}{2}\left(E_z^n + E_z^{n-1} + S_{E,1}^{n-1} - S_{E,2}^{n-1}\right). \tag{10}$$

Next, eliminate H_y using equations (1), (5), and (6). Applying $C_b \delta_x^H$ to both sides of equation (6) gives

$$C_b \delta_x^H H_y^n = C_b \delta_x^H H_y^{n-\frac{1}{2}} + \gamma \delta_x^{E,2} E_z^{n-\frac{1}{2}}. \tag{11}$$

Eliminate $C_b \delta_x^H H_y^{n-\frac{1}{2}}$ by equation (5) gives

$$C_b \delta_x^H H_y^n = E_z^n - \left(1-\gamma\delta_x^{E,2}\right) E_z^{n-\frac{1}{2}} - S_{E,2}^{n-1}. \tag{12}$$

Substituting into equation (1) to eliminate H_y:

$$\left(1-\gamma\delta_x^{E,2}\right)\left(E_z^{n+\frac{1}{2}} + E_z^{n-\frac{1}{2}}\right) = 2E_z^n + S_{E,1}^n - S_{E,2}^{n-1}. \tag{13}$$

Finally, substitute equations (9) and (10) into equation (13) to eliminate the half-step values of E_z:

$$\left(1-\gamma\delta_x^{E,2}\right) E_z^{n+1} = 2\left(1+\gamma\delta_x^{E,2}\right) E_z^n - \left(1-\gamma\delta_x^{E,2}\right) E_z^{n-1} + \left(1+\gamma\delta_x^{E,2}\right) S_{E,1}^n - \left(1-\gamma\delta_x^{E,2}\right) S_{E,1}^{n-1} + \left(1-\gamma\delta_x^{E,2}\right) S_{E,2}^n - \left(1+\gamma\delta_x^{E,2}\right) S_{E,2}^{n-1}. \tag{14}$$

Recall that $\gamma = C_b D_b = c^2 \Delta t^2 / 4$, then equation (14) results in,

$$\frac{E_z^{n+1} - 2E_z^n + E_z^{n-1}}{\Delta t^2} = \\ c^2 \delta_x^{E,2} \left(\frac{E_z^{n+1} + 2E_z^n + E_z^{n-1}}{4} \right) + \frac{1}{\Delta t^2} \left[\left(1 + \gamma \delta_x^{E,2}\right) S_{E,1}^n - \left(1 - \gamma \delta_x^{E,2}\right) S_{E,1}^{n-1} + \left(1 - \gamma \delta_x^{E,2}\right) S_{E,2}^n - \left(1 + \gamma \delta_x^{E,2}\right) S_{E,2}^{n-1} \right]. \tag{15}$$

Suppose the same source is applied in both substeps, with $S_{E,1}^n = S_{E,2}^n = \frac{\Delta t}{2} S^{n+\frac{1}{2}}$ and $S_{E,1}^{n-1} = S_{E,2}^{n-1} = \frac{\Delta t}{2} S^{n-\frac{1}{2}}$. Then equation (15) becomes

$$\frac{E_z^{n+1} - 2E_z^n + E_z^{n-1}}{\Delta t^2} = c^2 \delta_x^{E,2} \left(\frac{E_z^{n+1} + 2E_z^n + E_z^{n-1}}{4} \right) + \frac{S^{n+\frac{1}{2}} - S^{n-\frac{1}{2}}}{\Delta t}. \tag{16}$$

Assume $S^{n+a} = S(t_n + a\Delta t)$ and $F^n = S_t(t_n)$. The source approximation in equation (16) has the Taylor expansion

$$\frac{S^{n+\frac{1}{2}} - S^{n-\frac{1}{2}}}{\Delta t} = F^n + \frac{\Delta t^2}{24} S_{ttt}^n + O(\Delta t^4), \qquad \text{error} = \frac{\Delta t^2}{24} S_{ttt}^n + O(\Delta t^4). \tag{17}$$

Thus the midpoint-source case is second-order accurate for $F^n = S_t(t_n)$.

Alternatively, suppose the first source is evaluated at the first quarter-step and the second source at the third quarter-step: $S_{E,1}^n = \frac{\Delta t}{2} S^{n+\frac{1}{4}}$, $S_{E,2}^n = \frac{\Delta t}{2} S^{n+\frac{3}{4}}$, $S_{E,1}^{n-1} = \frac{\Delta t}{2} S^{n-\frac{3}{4}}$, and $S_{E,2}^{n-1} = \frac{\Delta t}{2} S^{n-\frac{1}{4}}$. Then equation (15) becomes

$$\frac{E_z^{n+1} - 2E_z^n + E_z^{n-1}}{\Delta t^2} = c^2 \delta_x^{E,2} \left(\frac{E_z^{n+1} + 2E_z^n + E_z^{n-1}}{4} \right) + \frac{1}{2\Delta t} \left[\left(1 + \gamma \delta_x^{E,2}\right) S^{n+\frac{1}{4}} - \left(1 - \gamma \delta_x^{E,2}\right) S^{n-\frac{3}{4}} + \left(1 - \gamma \delta_x^{E,2}\right) S^{n+\frac{3}{4}} - \left(1 + \gamma \delta_x^{E,2}\right) \right. \tag{18}$$

The source approximation in equation (18) has the Taylor expansion

$$F^n - \frac{\gamma}{2} \delta_x^{E,2} F^n + \frac{7\Delta t^2}{96} S_{ttt}^n + O(\Delta t^4), \qquad \text{error} = -\frac{\gamma}{2} \delta_x^{E,2} F^n + \frac{7\Delta t^2}{96} S_{ttt}^n + O(\Delta t^4). \tag{19}$$

Since $\gamma = O(\Delta t^2)$, the quarter-step source case is also second-order accurate.

Assume $S(x,t) = \widehat{S} e^{i(kx + \omega t)}$, and define $\theta = \omega \Delta t$, $\delta_x^{E,2} e^{ikx} = \lambda_x e^{ikx}$, and $q = -\gamma \lambda_x \geq 0$. Substituting into the source term on the right-hand side of equation (16) gives

$$\begin{aligned} F_{\text{mid}}^n &= \frac{S^{n+\frac{1}{2}} - S^{n-\frac{1}{2}}}{\Delta t} \\ &= \frac{\widehat{S} e^{i(kx + \omega t_n)}}{\Delta t} \left(e^{i\theta/2} - e^{-i\theta/2} \right) \\ &= \frac{2i\widehat{S}}{\Delta t} \sin\left(\frac{\theta}{2}\right) e^{i(kx + \omega t_n)}, \end{aligned} \tag{20}$$

so $\widehat{F}_{\text{mid}} = \frac{2i\widehat{S}}{\Delta t} \sin\left(\frac{\theta}{2}\right)$. Substituting into the source term on the right-hand side of equation (18) gives

$$\begin{aligned} F_{\text{quarter}}^n &= \frac{1}{2\Delta t} \left[\left(1 + \gamma \delta_x^{E,2}\right) S^{n+\frac{1}{4}} - \left(1 - \gamma \delta_x^{E,2}\right) S^{n-\frac{3}{4}} + \left(1 - \gamma \delta_x^{E,2}\right) S^{n+\frac{3}{4}} - \left(1 + \gamma \delta_x^{E,2}\right) S^{n-\frac{1}{4}} \right] \\ &= \frac{\widehat{S} e^{i(kx + \omega t_n)}}{2\Delta t} \left[(1-q) e^{i\theta/4} - (1+q) e^{-3i\theta/4} + (1+q) e^{3i\theta/4} - (1-q) e^{-i\theta/4} \right] \\ &= \frac{i\widehat{S}}{\Delta t} \left[(1-q) \sin\left(\frac{\theta}{4}\right) + (1+q) \sin\left(\frac{3\theta}{4}\right) \right] e^{i(kx + \omega t_n)}. \end{aligned} \tag{21}$$

so $\widehat{F}_{\text{quarter}} = \frac{i\widehat{S}}{\Delta t} \left[(1-q) \sin\left(\frac{\theta}{4}\right) + (1+q) \sin\left(\frac{3\theta}{4}\right) \right]$. Therefore the magnitude ratio between the quarter-step and midpoint-source terms is

$$\frac{\left|\widehat{F}_{\text{quarter}}\right|}{\left|\widehat{F}_{\text{mid}}\right|} = \left| \cos\left(\frac{\theta}{4}\right) + \frac{q \cos\left(\frac{\theta}{2}\right)}{2 \cos\left(\frac{\theta}{4}\right)} \right|. \tag{22}$$

For a spatially uniform source, $q = 0$, so the quarter-step magnitude is smaller by the factor

$$\frac{\left|\widehat{F}_{\text{quarter}}\right|}{\left|\widehat{F}_{\text{mid}}\right|} = \left| \cos\left(\frac{\omega \Delta t}{4}\right) \right|. \tag{23}$$

1.2 Magnetic field source terms

Let $S_{H,1}^{n}$ be the magnetic field source term in between n and $n+\frac{1}{2}$, and $S_{H,2}^{n}$ be that in between $n+\frac{1}{2}$ and $n+1$.

$$\left(1-\gamma\delta_x^{E,2}\right)E_z^{n+\frac{1}{2}} = E_z^n + C_b\delta_x^H H_y^n. \tag{24}$$

$$H_y^{n+\frac{1}{2}} = H_y^n + D_b\delta_x^E E_z^{n+\frac{1}{2}} + S_{H,1}^n. \tag{25}$$

$$E_z^{n+1} = E_z^{n+\frac{1}{2}} + C_b\delta_x^H H_y^{n+\frac{1}{2}}. \tag{26}$$

$$H_y^{n+1} = H_y^{n+\frac{1}{2}} + D_b\delta_x^E E_z^{n+\frac{1}{2}} + S_{H,2}^n. \tag{27}$$

Shifting back equations (26) and (27) one full step gives

$$E_z^n = E_z^{n-\frac{1}{2}} + C_b\delta_x^H H_y^{n-\frac{1}{2}}. \tag{28}$$

$$H_y^n = H_y^{n-\frac{1}{2}} + D_b\delta_x^E E_z^{n-\frac{1}{2}} + S_{H,2}^{n-1}. \tag{29}$$

To eliminate E_z from equations (25)–(27), equation (25) gives

$$D_b\delta_x^E E_z^{n+\frac{1}{2}} = H_y^{n+\frac{1}{2}} - H_y^n - S_{H,1}^n. \tag{30}$$

Substituting into equation (27) gives

$$H_y^{n+1} = 2H_y^{n+\frac{1}{2}} - H_y^n - S_{H,1}^n + S_{H,2}^n. \tag{31}$$

Solving for the half-step value gives

$$H_y^{n+\frac{1}{2}} = \frac{1}{2}\left(H_y^{n+1} + H_y^n + S_{H,1}^n - S_{H,2}^n\right). \tag{32}$$

Shifting back one full step gives

$$H_y^{n-\frac{1}{2}} = \frac{1}{2}\left(H_y^n + H_y^{n-1} + S_{H,1}^{n-1} - S_{H,2}^{n-1}\right). \tag{33}$$

Next, eliminate E_z using equations (24), (25), (28), and (29). Applying $D_b\delta_x^E$ to both sides of equation (28) gives

$$D_b\delta_x^E E_z^n = D_b\delta_x^E E_z^{n-\frac{1}{2}} + \gamma\delta_x^{H,2} H_y^{n-\frac{1}{2}}. \tag{34}$$

Eliminate $D_b\delta_x^E E_z^{n-\frac{1}{2}}$ by equation (29) gives

$$D_b\delta_x^E E_z^n = H_y^n - \left(1-\gamma\delta_x^{H,2}\right)H_y^{n-\frac{1}{2}} - S_{H,2}^{n-1}. \tag{35}$$

Applying $D_b\delta_x^E$ to both sides of equation (24) gives

$$\left(1-\gamma\delta_x^{H,2}\right)D_b\delta_x^E E_z^{n+\frac{1}{2}} = D_b\delta_x^E E_z^n + \gamma\delta_x^{H,2} H_y^n. \tag{36}$$

Eliminate $D_b\delta_x^E E_z^{n+\frac{1}{2}}$ by equation (25) gives

$$D_b\delta_x^E E_z^n = \left(1-\gamma\delta_x^{H,2}\right)H_y^{n+\frac{1}{2}} - H_y^n - \left(1-\gamma\delta_x^{H,2}\right)S_{H,1}^n. \tag{37}$$

Equating equations (37) and (35) eliminates E_z^n:

$$\left(1-\gamma\delta_x^{H,2}\right)\left(H_y^{n+\frac{1}{2}} + H_y^{n-\frac{1}{2}}\right) = 2H_y^n + \left(1-\gamma\delta_x^{H,2}\right)S_{H,1}^n - S_{H,2}^{n-1}. \tag{38}$$

Finally, substitute equations (32) and (33) into equation (38) to eliminate the half-step values of H_y:

$$\left(1-\gamma\delta_x^{H,2}\right)H_y^{n+1} = 2\left(1+\gamma\delta_x^{H,2}\right)H_y^n - \left(1-\gamma\delta_x^{H,2}\right)H_y^{n-1} + \left(1-\gamma\delta_x^{H,2}\right)S_{H,1}^n - \left(1-\gamma\delta_x^{H,2}\right)S_{H,1}^{n-1} + \left(1-\gamma\delta_x^{H,2}\right)S_{H,2}^n - \left(1+\gamma\delta_x^{H,2}\right)S_{H,2}^{n-1}. \tag{39}$$

Recall that $\gamma = C_bD_b = c^2\Delta t^2/4$, then equation (39) results in,

$$\begin{aligned}&\frac{H_y^{n+1} - 2H_y^n + H_y^{n-1}}{\Delta t^2} = \\ &c^2\delta_x^{H,2}\left(\frac{H_y^{n+1} + 2H_y^n + H_y^{n-1}}{4}\right) + \frac{1}{\Delta t^2}\left[\left(1-\gamma\delta_x^{H,2}\right)S_{H,1}^n - \left(1-\gamma\delta_x^{H,2}\right)S_{H,1}^{n-1} + \left(1-\gamma\delta_x^{H,2}\right)S_{H,2}^n - \left(1+\gamma\delta_x^{H,2}\right)S_{H,2}^{n-1}\right].\end{aligned} \tag{40}$$

Suppose the source is added only once in the first substep and evaluated at the midpoint, with $S^n_{H,1} = \Delta t S_H^{n+\frac{1}{2}}$, $S^n_{H,2} = 0$, $S^{n-1}_{H,1} = \Delta t S_H^{n-\frac{1}{2}}$, and $S^{n-1}_{H,2} = 0$. Then the source terms in equation (40) reduce to

$$\Delta t \left(1 - \gamma \delta_x^{H,2}\right) \left(S_H^{n+\frac{1}{2}} - S_H^{n-\frac{1}{2}}\right). \tag{41}$$

Therefore equation (40) becomes

$$\frac{H_y^{n+1} - 2H_y^n + H_y^{n-1}}{\Delta t^2} = c^2 \delta_x^{H,2} \left(\frac{H_y^{n+1} + 2H_y^n + H_y^{n-1}}{4}\right) + \left(1 - \gamma \delta_x^{H,2}\right) \frac{S_H^{n+\frac{1}{2}} - S_H^{n-\frac{1}{2}}}{\Delta t}. \tag{42}$$

Assume $S_H^{n+a} = S_H(t_n + a\Delta t)$ and $F_H^n = (S_H)_t(t_n)$. The source approximation in equation (42) has the Taylor expansion

$$F_H^n - \gamma \delta_x^{H,2} F_H^n + \frac{\Delta t^2}{24} (S_H)^n_{ttt} + O(\Delta t^4). \tag{43}$$

Since $\gamma = O(\Delta t^2)$, the single midpoint-source case is second-order accurate for $F_H^n = (S_H)_t(t_n)$.

Alternatively, suppose the first source is evaluated at the first quarter-step and the second source at the third quarter-step: $S^n_{H,1} = \frac{\Delta t}{2} S_H^{n+\frac{1}{4}}$, $S^n_{H,2} = \frac{\Delta t}{2} S_H^{n+\frac{3}{4}}$, $S^{n-1}_{H,1} = \frac{\Delta t}{2} S_H^{n-\frac{3}{4}}$, and $S^{n-1}_{H,2} = \frac{\Delta t}{2} S_H^{n-\frac{1}{4}}$. Then equation (40) becomes

$$\begin{aligned} &\frac{H_y^{n+1} - 2H_y^n + H_y^{n-1}}{\Delta t^2} = \\ &c^2 \delta_x^{H,2} \left(\frac{H_y^{n+1} + 2H_y^n + H_y^{n-1}}{4}\right) + \frac{1}{2\Delta t} \left[\left(1 - \gamma \delta_x^{H,2}\right) S_H^{n+\frac{1}{4}} - \left(1 - \gamma \delta_x^{H,2}\right) S_H^{n-\frac{3}{4}} + \left(1 - \gamma \delta_x^{H,2}\right) S_H^{n+\frac{3}{4}} - \left(1 + \gamma \delta_x^{H,2}\right) S_H^{n-\frac{1}{4}}\right]. \end{aligned} \tag{44}$$

The source approximation in equation (44) has the Taylor expansion

$$F_H^n - \frac{\gamma}{\Delta t} \delta_x^{H,2} S_H^n - \frac{3\gamma}{4} \delta_x^{H,2} F_H^n + \frac{7\Delta t^2}{96} (S_H)^n_{ttt} + O(\Delta t^3). \tag{45}$$

Thus the quarter-step source case is however first-order accurate for a spatially varying S_H, and second-order when $\delta_x^{H,2} S_H = 0$.

Assume $S_H(x,t) = \widehat{S}_H e^{i(kx + \omega t)}$, and define $\theta = \omega \Delta t$, $\delta_x^{H,2} e^{ikx} = \lambda_{H,x} e^{ikx}$, and $q_H = -\gamma \lambda_{H,x} \geq 0$. Substituting into the source term on the right-hand side of equation (42) gives

$$\begin{aligned} F^n_{H,\text{mid}} &= \left(1 - \gamma \delta_x^{H,2}\right) \frac{S_H^{n+\frac{1}{2}} - S_H^{n-\frac{1}{2}}}{\Delta t} \\ &= \frac{\left(1 + q_H\right) \widehat{S}_H e^{i(kx + \omega t_n)}}{\Delta t} \left(e^{i\theta/2} - e^{-i\theta/2}\right) \\ &= \frac{2i\left(1 + q_H\right) \widehat{S}_H}{\Delta t} \sin\left(\frac{\theta}{2}\right) e^{i(kx + \omega t_n)}. \end{aligned} \tag{46}$$

so $\widehat{F}_{H,\text{mid}} = \frac{2i(1+q_H)\widehat{S}_H}{\Delta t} \sin\left(\frac{\theta}{2}\right)$. Substituting into the source term on the right-hand side of equation (44) gives

$$\begin{aligned} F^n_{H,\text{quarter}} &= \frac{1}{2\Delta t} \left[\left(1 - \gamma \delta_x^{H,2}\right) S_H^{n+\frac{1}{4}} - \left(1 - \gamma \delta_x^{H,2}\right) S_H^{n-\frac{3}{4}} + \left(1 - \gamma \delta_x^{H,2}\right) S_H^{n+\frac{3}{4}} - \left(1 + \gamma \delta_x^{H,2}\right) S_H^{n-\frac{1}{4}}\right] \\ &= \frac{\widehat{S}_H e^{i(kx + \omega t_n)}}{2\Delta t} \left[\left(1 + q_H\right) e^{i\theta/4} - \left(1 + q_H\right) e^{-3i\theta/4} + \left(1 + q_H\right) e^{3i\theta/4} - \left(1 - q_H\right) e^{-i\theta/4}\right] \\ &= \frac{\widehat{S}_H}{\Delta t} \left[q_H \cos\left(\frac{\theta}{4}\right) + i\left(\sin\left(\frac{\theta}{4}\right) + \left(1 + q_H\right) \sin\left(\frac{3\theta}{4}\right)\right)\right] e^{i(kx + \omega t_n)}. \end{aligned} \tag{47}$$

Therefore the magnitude ratio between the quarter-step and midpoint magnetic-source terms is

$$\frac{\left|\widehat{F}_{H,\text{quarter}}\right|}{\left|\widehat{F}_{H,\text{mid}}\right|} = \frac{\sqrt{q_H^2 \cos^2\left(\frac{\theta}{4}\right) + \left[\sin\left(\frac{\theta}{4}\right) + \left(1 + q_H\right) \sin\left(\frac{3\theta}{4}\right)\right]^2}}{2\left(1 + q_H\right) \left|\sin\left(\frac{\theta}{2}\right)\right|}. \tag{48}$$

For a spatially uniform source, $q_H = 0$, this reduces to

$$\frac{\left|\widehat{F}_{H,\text{quarter}}\right|}{\left|\widehat{F}_{H,\text{mid}}\right|} = \left|\cos\left(\frac{\omega \Delta t}{4}\right)\right|. \tag{49}$$